Chemical Adventure

AN INTERACTIVE CHEMISTRY LEARNING GAME (REACT + NODE.JS + MONGODB)

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Project Overview

- Periodic tabel game for students (Periodic Table, Quiz, Molecules)
- Full-stack app: React + Node.js + MongoDB Atlas
- Design UI : Figma
- CSS Animation , JavaScript
- DOM Manipulation (JS)

Folder Structure

```
chemical-adventure/
│
├── public/
│   └── index.html
├── src/
│   ├── components/
│   │   ├── PeriodicTable.jsx
│   │   ├── MoleculeMixer.jsx
│   │   ├── AtomQuiz.jsx
│   │   ├── ChemicalReaction.jsx
│   │   └── ScoreBoard.jsx
│   ├── data/
│   │   └── elements.js
│   ├── styles/
│   │   ├── main.css
│   │   ├── quiz.css
│   │   └── reaction.css
│   ├── App.jsx
│   └── index.js
├── backend/
│   └── server.js
├── README.me
```

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta http-equiv="X-UA-Compatible" content="IE=edge" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>Chemical Adventure</title>
  <link rel="stylesheet" href="style.css" />
  <link rel="icon" href="%PUBLIC_URL%/favicon.ico" />
  <meta name="description" content="Interactive chemistry learning platform
with periodic table and quizzes." />
  <meta name="author" content="Chemical Adventure Team" />
</head>
```

```
<body>
  <div id="root">
    <header>
      <h1>Chemical Adventure</h1>
      <nav>
        <ul>
          <li><a href="#table">Periodic Table</a></li>
          <li><a href="#quiz">Atom Quiz</a></li>
          <li><a href="#mixer">Molecule Mixer</a></li>
          <li><a href="#scoreboard">Scoreboard</a></li>
        </ul>
      </nav>
    </header>
```

```
<main>
    <section id="table">
      <h2>Periodic Table</h2>
      <div class="periodic-table">
        <div class="element">H</div><div
class="element">He</div><div
class="element">Li</div>
        <div class="element">Be</div><div
class="element">B</div><div
class="element">C</div>
        <div class="element">N</div><div
class="element">O</div><div
class="element">F</div>
        <div class="element">Ne</div>
      </div>
    </section>
```

```
 <section id="mixer">
        <h2>Molecule Mixer</h2>
        <p>Combine elements to create molecules.</p>
        <button>H</button><button>O</button><button>Cl</button>
        <div id="result">Result will appear here</div>
      </section>
      <section id="scoreboard">
        <h2>Scoreboard</h2>
        <ul>
          <li>User1: 85</li>
          <li>User2: 78</li>
          <li>User3: 90</li>
        </ul>
      </section>
    </main>
    <footer>
      <p>&copy; 2025 Chemical Adventure. All rights reserved.</p>
    </footer>
  </div>
</body>
</html>
```

Style.css

```
body {
  margin: 0;
  padding: 0;
  background-color: #121212;
  color: #f0f0f0;
  font-family: Arial, sans-serif;
}

header {
  background-color: #1e1e1e;
  padding: 20px;
  text-align: center;
}

nav ul {
  list-style: none;
  padding: 0;
  display: flex;
  justify-content: center;
  gap: 15px;
}
```

```css
nav a {
  color: #00ffd0;
  text-decoration: none;
  font-weight: bold;
}

main {
  padding: 20px;
}

.periodic-table {
  display: grid;
  grid-template-columns: repeat(10, 1fr);
  gap: 10px;
  margin-top: 10px;
}

.element {
  background-color: #222;
  padding: 10px;
  border-radius: 4px;
  text-align: center;
  font-weight: bold;
  border: 1px solid #00ffd0;
}
```

```css
button {
  background-color: #00ffd0;
  border: none;
  color: #000;
  padding: 10px 15px;
  margin: 5px;
  cursor: pointer;
  font-weight: bold;
  transition: background-color 0.3s ease;
}

button:hover {
  background-color: #00ccaa;
}

#scoreboard ul {
  padding-left: 20px;
}

footer {
  text-align: center;
  padding: 10px;
  background-color: #1e1e1e;
  color: #aaa;
  position: relative;
  bottom: 0;
  width: 100%;
}
```

index.js

```
import React, { Suspense } from 'react';
import ReactDOM from 'react-dom/client';
import { BrowserRouter as Router, Routes, Route, Navigate } from 'react-
router-dom';
import { Provider } from 'react-redux';
import store from './redux/store';
import './styles/main.css';
import App from './App';
import PeriodicTable from './components/PeriodicTable';
import AtomQuiz from './components/AtomQuiz';
import MoleculeMixer from './components/MoleculeMixer';
import ScoreBoard from './components/ScoreBoard';
import { ThemeProvider } from './context/ThemeContext';
import { UserProvider } from './context/UserContext';

const LazyAbout = React.lazy(() => import('./components/About'));
const LazyHelp = React.lazy(() => import('./components/Help'));
const LazyLogin = React.lazy(() => import('./components/Login'));
const LazyRegister = React.lazy(() => import('./components/Register'));
const LazyProfile = React.lazy(() => import('./components/Profile'));
const LazySettings = React.lazy(() => import('./components/Settings'));
const LazyDashboard = React.lazy(() => import('./components/Dashboard'));
```

```
function ErrorFallback() {
  return (
    <div style={{
      color: 'red',
      padding: '2rem',
      fontWeight: 'bold',
      textAlign: 'center'
    }}>
      Something went wrong while loading the page.
    </div>
  );
}

function AuthGuard({ children }) {
  const isAuthenticated = localStorage.getItem('authToken');
  return isAuthenticated ? children : <Navigate to="/login" />;
}

function RootApp() {
  const visited = localStorage.getItem('visited');
  if (!visited) {
    localStorage.setItem('visited', 'true');
    console.log('Welcome to Chemical Adventure!');
  }
```

```
return (
  <Provider store={store}>
    <ThemeProvider>
      <UserProvider>
        <Router>
          <Suspense fallback={<div className="loading-screen">Loading...</div>}>
            <Routes>
              <Route path="/" element={<App />} />
              <Route path="/table" element={<PeriodicTable />} />
              <Route path="/quiz" element={<AtomQuiz />} />
              <Route path="/mixer" element={<MoleculeMixer />} />
              <Route path="/scoreboard" element={<ScoreBoard />} />
              <Route path="/about" element={<LazyAbout />} />
              <Route path="/help" element={<LazyHelp />} />
              <Route path="/login" element={<LazyLogin />} />
              <Route path="/register" element={<LazyRegister />} />
              <Route path="/dashboard" element={
                <AuthGuard><LazyDashboard /></AuthGuard>
              } />
              <Route path="/profile" element={
                <AuthGuard><LazyProfile /></AuthGuard>
              } />
              <Route path="/settings" element={
                <AuthGuard><LazySettings /></AuthGuard>
              } />
              <Route path="*" element={
                <div style={{ padding: '20px', textAlign: 'center' }}>
                  <h2>404 - Page Not Found</h2>
                </div>
              } />
            </Routes>
          </Suspense>
        </Router>
      </UserProvider>
    </ThemeProvider>
  </Provider>
 );
}

const root = ReactDOM.createRoot(document.getElementById('root'));
root.render(<RootApp />);
```

App.jsx

```
import React, { useState, useEffect, useContext } from 'react';
import { Link, useNavigate, useLocation } from 'react-router-dom';
import PeriodicTable from './components/PeriodicTable';
import AtomQuiz from './components/AtomQuiz';
import MoleculeMixer from './components/MoleculeMixer';
import ScoreBoard from './components/ScoreBoard';
import { ThemeContext } from './context/ThemeContext';

function App() {
  const { theme, toggleTheme } = useContext(ThemeContext);
  const [activeSection, setActiveSection] = useState('home');
  const [showWelcome, setShowWelcome] = useState(true);
  const navigate = useNavigate();
  const location = useLocation();

  useEffect(() => {
    // Hide welcome message after 5 seconds
    const timer = setTimeout(() => {
      setShowWelcome(false);
    }, 5000);
```

```
import AtomQuiz from './components/AtomQuiz';
import ChemicalReaction from
'./components/ChemicalReaction';

function App() {
  return (
    <div>
      <PeriodicTable />
      <MoleculeMixer />
      <AtomQuiz />
      <ChemicalReaction />
      <ScoreBoard />
    </div>
  );
}
```

```
const path = location.pathname;
    if (path.includes('table')) setActiveSection('table');
    else if (path.includes('quiz')) setActiveSection('quiz');
    else if (path.includes('mixer')) setActiveSection('mixer');
    else if (path.includes('scoreboard')) setActiveSection('scoreboard');
    else setActiveSection('home');

    return () => clearTimeout(timer);
  }, [location]);

  function handleNavigate(section) {
    setActiveSection(section);
    switch (section) {
      case 'table':
        navigate('/table');
        break;
      case 'quiz':
        navigate('/quiz');
        break;
      case 'mixer':
        navigate('/mixer');
        break;
      case 'scoreboard':
        navigate('/scoreboard');
        break;
      default:
        navigate('/');
    }
  }

  return (
    <div className={`app-container ${theme}`}>
      <header className="app-header">
        <h1 className="app-title">Chemical Adventure</h1>
        <button className="theme-toggle-btn" onClick={toggleTheme}>
          Switch to {theme === 'dark' ? 'Light' : 'Dark'} Mode
        </button>
      </header>
```

```
{showWelcome && (
      <section className="welcome-banner">
        <h2>Welcome to Chemical Adventure!</h2>
        <p>Explore all 118 elements, take quizzes, and mix molecules.</p>
      </section>
    )}

    <nav className="app-nav">
      <ul>
        <li
          className={activeSection === 'home' ? 'active' : ''}
          onClick={() => handleNavigate('home')}
        >
          Home
        </li>
        <li
          className={activeSection === 'table' ? 'active' : ''}
          onClick={() => handleNavigate('table')}
        >
          Periodic Table
        </li>
        <li
          className={activeSection === 'quiz' ? 'active' : ''}
          onClick={() => handleNavigate('quiz')}
        >
          Atom Quiz
        </li>
        <li
          className={activeSection === 'mixer' ? 'active' : ''}
          onClick={() => handleNavigate('mixer')}
        >
          Molecule Mixer
        </li>
        <li
          className={activeSection === 'scoreboard' ? 'active' : ''}
          onClick={() => handleNavigate('scoreboard')}
        >
          Scoreboard
        </li>
      </ul>
    </nav>
```

```
 <main className="app-main">
       {activeSection === 'home' && (
         <section className="home-section">
           <h2>About Chemical Adventure</h2>
</section>
       )}

       {activeSection === 'table' && <PeriodicTable />}

       {activeSection === 'quiz' && <AtomQuiz />}

       {activeSection === 'mixer' && <MoleculeMixer />}

       {activeSection === 'scoreboard' && <ScoreBoard />}
     </main>

     <footer className="app-footer">
       <p>&copy; 2025 Chemical Adventure. All rights reserved.</p>
       <small>
         Developed by Your Name. Powered by React and MongoDB Atlas backend.
       </small>
     </footer>
   </div>
  );
}

export default App;
```

PeriodicTable.jsx

```
import React, { useState } from 'react';

const elements = [
  { number: 1, symbol: 'H', name: 'Hydrogen', group: 1, period: 1 },
  { number: 2, symbol: 'He', name: 'Helium', group: 18, period: 1 },
  { number: 3, symbol: 'Li', name: 'Lithium', group: 1, period: 2 },
  { number: 4, symbol: 'Be', name: 'Beryllium', group: 2, period: 2 },
  { number: 5, symbol: 'B', name: 'Boron', group: 13, period: 2 },
  { number: 6, symbol: 'C', name: 'Carbon', group: 14, period: 2 },
  { number: 7, symbol: 'N', name: 'Nitrogen', group: 15, period: 2 },
  { number: 8, symbol: 'O', name: 'Oxygen', group: 16, period: 2 },
  { number: 9, symbol: 'F', name: 'Fluorine', group: 17, period: 2 },
  { number: 10, symbol: 'Ne', name: 'Neon', group: 18, period: 2 },
  { number: 11, symbol: 'Na', name: 'Sodium', group: 1, period: 3 },
  { number: 12, symbol: 'Mg', name: 'Magnesium', group: 2, period: 3 },
  { number: 13, symbol: 'Al', name: 'Aluminum', group: 13, period: 3 },
  { number: 14, symbol: 'Si', name: 'Silicon', group: 14, period: 3 },
  { number: 15, symbol: 'P', name: 'Phosphorus', group: 15, period: 3 },
  { number: 16, symbol: 'S', name: 'Sulfur', group: 16, period: 3 },
  { number: 17, symbol: 'Cl', name: 'Chlorine', group: 17, period: 3 },
  { number: 18, symbol: 'Ar', name: 'Argon', group: 18, period: 3 },
  { number: 19, symbol: 'K', name: 'Potassium', group: 1, period: 4 },
  { number: 20, symbol: 'Ca', name: 'Calcium', group: 2, period: 4 },
  { number: 21, symbol: 'Sc', name: 'Scandium', group: 3, period: 4 },
  { number: 22, symbol: 'Ti', name: 'Titanium', group: 4, period: 4 },
  { number: 23, symbol: 'V', name: 'Vanadium', group: 5, period: 4 },
  { number: 24, symbol: 'Cr', name: 'Chromium', group: 6, period: 4 },
  { number: 25, symbol: 'Mn', name: 'Manganese', group: 7, period: 4 },
  { number: 26, symbol: 'Fe', name: 'Iron', group: 8, period: 4 },
  { number: 27, symbol: 'Co', name: 'Cobalt', group: 9, period: 4 },
```

```
{ number: 28, symbol: 'Ni', name: 'Nickel', group: 10, period: 4 },
  { number: 29, symbol: 'Cu', name: 'Copper', group: 11, period: 4 },
  { number: 30, symbol: 'Zn', name: 'Zinc', group: 12, period: 4 },
  { number: 31, symbol: 'Ga', name: 'Gallium', group: 13, period: 4 },
  { number: 32, symbol: 'Ge', name: 'Germanium', group: 14, period: 4 },
  { number: 33, symbol: 'As', name: 'Arsenic', group: 15, period: 4 },
  { number: 34, symbol: 'Se', name: 'Selenium', group: 16, period: 4 },
  { number: 35, symbol: 'Br', name: 'Bromine', group: 17, period: 4 },
  { number: 36, symbol: 'Kr', name: 'Krypton', group: 18, period: 4 },
  { number: 37, symbol: 'Rb', name: 'Rubidium', group: 1, period: 5 },
  { number: 38, symbol: 'Sr', name: 'Strontium', group: 2, period: 5 },
  { number: 39, symbol: 'Y', name: 'Yttrium', group: 3, period: 5 },
  { number: 40, symbol: 'Zr', name: 'Zirconium', group: 4, period: 5 },
  { number: 41, symbol: 'Nb', name: 'Niobium', group: 5, period: 5 },
  { number: 42, symbol: 'Mo', name: 'Molybdenum', group: 6, period: 5 },
  { number: 43, symbol: 'Tc', name: 'Technetium', group: 7, period: 5 },
  { number: 44, symbol: 'Ru', name: 'Ruthenium', group: 8, period: 5 },
  { number: 45, symbol: 'Rh', name: 'Rhodium', group: 9, period: 5 },
  { number: 46, symbol: 'Pd', name: 'Palladium', group: 10, period: 5 },
  { number: 47, symbol: 'Ag', name: 'Silver', group: 11, period: 5 },
  { number: 48, symbol: 'Cd', name: 'Cadmium', group: 12, period: 5 },
  { number: 49, symbol: 'In', name: 'Indium', group: 13, period: 5 },
  { number: 50, symbol: 'Sn', name: 'Tin', group: 14, period: 5 },
  { number: 51, symbol: 'Sb', name: 'Antimony', group: 15, period: 5 },
  { number: 52, symbol: 'Te', name: 'Tellurium', group: 16, period: 5 },
  { number: 53, symbol: 'I', name: 'Iodine', group: 17, period: 5 },
  { number: 54, symbol: 'Xe', name: 'Xenon', group: 18, period: 5 },
  { number: 55, symbol: 'Cs', name: 'Cesium', group: 1, period: 6 },
  { number: 56, symbol: 'Ba', name: 'Barium', group: 2, period: 6 },
  { number: 57, symbol: 'La', name: 'Lanthanum', group: 3, period: 6 },
  { number: 58, symbol: 'Ce', name: 'Cerium', group: null, period: 6 },
  { number: 59, symbol: 'Pr', name: 'Praseodymium', group: null, period: 6 },
  { number: 60, symbol: 'Nd', name: 'Neodymium', group: null, period: 6 },
```

```
{ number: 61, symbol: 'Pm', name: 'Promethium', group: null, period: 6 },
 { number: 62, symbol: 'Sm', name: 'Samarium', group: null, period: 6 },
 { number: 63, symbol: 'Eu', name: 'Europium', group: null, period: 6 },
 { number: 64, symbol: 'Gd', name: 'Gadolinium', group: null, period: 6 },
 { number: 65, symbol: 'Tb', name: 'Terbium', group: null, period: 6 },
 { number: 66, symbol: 'Dy', name: 'Dysprosium', group: null, period: 6 },
 { number: 67, symbol: 'Ho', name: 'Holmium', group: null, period: 6 },
 { number: 68, symbol: 'Er', name: 'Erbium', group: null, period: 6 },
 { number: 69, symbol: 'Tm', name: 'Thulium', group: null, period: 6 },
 { number: 70, symbol: 'Yb', name: 'Ytterbium', group: null, period: 6 },
 { number: 71, symbol: 'Lu', name: 'Lutetium', group: 3, period: 6 },
 { number: 72, symbol: 'Hf', name: 'Hafnium', group: 4, period: 6 },
 { number: 73, symbol: 'Ta', name: 'Tantalum', group: 5, period: 6 },
 { number: 74, symbol: 'W', name: 'Tungsten', group: 6, period: 6 },
 { number: 75, symbol: 'Re', name: 'Rhenium', group: 7, period: 6 },
 { number: 76, symbol: 'Os', name: 'Osmium', group: 8, period: 6 },
 { number: 77, symbol: 'Ir', name: 'Iridium', group: 9, period: 6 },
 { number: 78, symbol: 'Pt', name: 'Platinum', group: 10, period: 6 },
 { number: 79, symbol: 'Au', name: 'Gold', group: 11, period: 6 },
 { number: 80, symbol: 'Hg', name: 'Mercury', group: 12, period: 6 },
 { number: 81, symbol: 'Tl', name: 'Thallium', group: 13, period: 6 },
 { number: 82, symbol: 'Pb', name: 'Lead', group: 14, period: 6 },
 { number: 83, symbol: 'Bi', name: 'Bismuth', group: 15, period: 6 },
 { number: 84, symbol: 'Po', name: 'Polonium', group: 16, period: 6 },
 { number: 85, symbol: 'At', name: 'Astatine', group: 17, period: 6 },
 { number: 86, symbol: 'Rn', name: 'Radon', group: 18, period: 6 },
 { number: 87, symbol: 'Fr', name: 'Francium', group: 1, period: 7 },
 { number: 88, symbol: 'Ra', name: 'Radium', group: 2, period: 7 },
 { number: 89, symbol: 'Ac', name: 'Actinium', group: 3, period: 7 },
 { number: 90, symbol: 'Th', name: 'Thorium', group: null, period: 7 },
 { number: 91, symbol: 'Pa', name: 'Protactinium', group: null, period: 7 },
 { number: 92, symbol: 'U', name: 'Uranium', group: null, period: 7 },
 { number: 93, symbol: 'Np', name: 'Neptunium', group: null, period: 7 },
 { number: 94, symbol: 'Pu', name: 'Plutonium', group: null, period: 7 },
 { number: 95, symbol: 'Am', name: 'Americium', group: null, period: 7 },
 { number: 96, symbol: 'Cm', name: 'Curium', group: null, period: 7 },
 { number: 97, symbol: 'Bk', name: 'Berkelium', group: null, period: 7 },
 { number: 98, symbol: 'Cf', name: 'Californium', group: null, period: 7 },
 { number: 99, symbol: 'Es', name: 'Einsteinium', group: null, period: 7 },
 { number: 100, symbol: 'Fm', name: 'Fermium', group: null, period: 7 },
 { number: 101, symbol: 'Md', name: 'Mendelevium', group: null, period: 7 },
 { number: 102, symbol: 'No', name: 'Nobelium', group: null, period: 7 },
 { number: 103, symbol: 'Lr', name: 'Lawrencium', group: 3, period: 7 },
 { number: 104, symbol: 'Rf', name: 'Rutherfordium', group: 4, period: 7 },
```

```
{ number: 105, symbol: 'Db', name: 'Dubnium', group: 5, period: 7 },
  { number: 106, symbol: 'Sg', name: 'Seaborgium', group: 6, period: 7 },
  { number: 107, symbol: 'Bh', name: 'Bohrium', group: 7, period: 7 },
  { number: 108, symbol: 'Hs', name: 'Hassium', group: 8, period: 7 },
  { number: 109, symbol: 'Mt', name: 'Meitnerium', group: 9, period: 7 },
  { number: 110, symbol: 'Ds', name: 'Darmstadtium', group: 10, period: 7 },
  { number: 111, symbol: 'Rg', name: 'Roentgenium', group: 11, period: 7 },
  { number: 112, symbol: 'Cn', name: 'Copernicium', group: 12, period: 7 },
  { number: 113, symbol: 'Nh', name: 'Nihonium', group: 13, period: 7 },
  { number: 114, symbol: 'Fl', name: 'Flerovium', group: 14, period: 7 },
  { number: 115, symbol: 'Mc', name: 'Moscovium', group: 15, period: 7 },
  { number: 116, symbol: 'Lv', name: 'Livermorium', group: 16, period: 7 },
  { number: 117, symbol: 'Ts', name: 'Tennessine', group: 17, period: 7 },
  { number: 118, symbol: 'Og', name: 'Oganesson', group: 18, period: 7 },
];

function PeriodicTable() {
  const [selectedElement, setSelectedElement] = useState(null);

  function handleElementClick(el) {
    setSelectedElement(el);
  }

  return (
    <div className="periodic-table-container">
      <h2>Periodic Table</h2>
      <div className="periodic-table-grid">
        {elements.map((el) => (
          <div
            key={el.number}
            className={`element-box group-${el.group ?? 'lanth-act'} period-${el.period}`}
            onClick={() => handleElementClick(el)}
            title={`${el.name} (Atomic Number: ${el.number})`}
```

```
style={{
          cursor: 'pointer',
          border: selectedElement?.number === el.number ? '3px solid #00ffd0' : '1px solid #555',
          padding: '10px',
          margin: '3px',
          borderRadius: '6px',
          backgroundColor: selectedElement?.number === el.number ? '#003333' : '#111',
          color: '#eee',
          textAlign: 'center',
          userSelect: 'none',
          minWidth: '50px',
        }}
      >
        <div style={{ fontSize: '0.7rem' }}>{el.number}</div>
        <div style={{ fontSize: '1.5rem', fontWeight: 'bold' }}>{el.symbol}</div>
        <div style={{ fontSize: '0.75rem' }}>{el.name}</div>
      </div>
    ))}
  </div>

  {selectedElement && (
    <div
      style={{
        marginTop: '20px',
        padding: '15px',
        backgroundColor: '#222',
        borderRadius: '8px',
        color: '#00ffd0',
        maxWidth: '350px',
      }}
```

```
 >
          <h3>{selectedElement.name} ({selectedElement.symbol})</h3>
          <p>Atomic Number: {selectedElement.number}</p>
          <p>Group: {selectedElement.group ?? 'N/A'}</p>
          <p>Period: {selectedElement.period}</p>
          <button
            onClick={() => setSelectedElement(null)}
            style={{
              backgroundColor: '#00ffd0',
              border: 'none',
              padding: '8px 12px',
              borderRadius: '4px',
              cursor: 'pointer',
              fontWeight: 'bold',
              color: '#000',
              marginTop: '10px',
            }}
          >
            Close
          </button>
        </div>
      )}
    </div>
  );
}

export default PeriodicTable;
```

AtomQuiz.jsx

```
import React, { useState, useEffect } from 'react';

const elements = [
  { number: 1, symbol: 'H', name: 'Hydrogen' },
  { number: 2, symbol: 'He', name: 'Helium' },
  { number: 3, symbol: 'Li', name: 'Lithium' },
  { number: 4, symbol: 'Be', name: 'Beryllium' },
  { number: 5, symbol: 'B', name: 'Boron' },
  { number: 6, symbol: 'C', name: 'Carbon' },
  { number: 7, symbol: 'N', name: 'Nitrogen' },
  { number: 8, symbol: 'O', name: 'Oxygen' },
  { number: 9, symbol: 'F', name: 'Fluorine' },
  { number: 10, symbol: 'Ne', name: 'Neon' },
  { number: 11, symbol: 'Na', name: 'Sodium' },
  { number: 12, symbol: 'Mg', name: 'Magnesium' },
  { number: 13, symbol: 'Al', name: 'Aluminum' },
  { number: 14, symbol: 'Si', name: 'Silicon' },
  { number: 15, symbol: 'P', name: 'Phosphorus' },
  { number: 16, symbol: 'S', name: 'Sulfur' },
  { number: 17, symbol: 'Cl', name: 'Chlorine' },
  { number: 18, symbol: 'Ar', name: 'Argon' },
  { number: 19, symbol: 'K', name: 'Potassium' },
  { number: 20, symbol: 'Ca', name: 'Calcium' },
```

```
{ number: 21, symbol: "Sc", name: "Scandium" },
 { number: 22, symbol: "Ti", name: "Titanium" },
 { number: 23, symbol: "V", name: "Vanadium" },
 { number: 24, symbol: "Cr", name: "Chromium" },
 { number: 25, symbol: "Mn", name: "Manganese" },
 { number: 26, symbol: "Fe", name: "Iron" },
 { number: 27, symbol: "Co", name: "Cobalt" },
 { number: 28, symbol: "Ni", name: "Nickel" },
 { number: 29, symbol: "Cu", name: "Copper" },
 { number: 30, symbol: "Zn", name: "Zinc" },
 { number: 31, symbol: "Ga", name: "Gallium" },
 { number: 32, symbol: "Ge", name: "Germanium" },
 { number: 33, symbol: "As", name: "Arsenic" },
 { number: 34, symbol: "Se", name: "Selenium" },
 { number: 35, symbol: "Br", name: "Bromine" },
 { number: 36, symbol: "Kr", name: "Krypton" },
 { number: 37, symbol: "Rb", name: "Rubidium" },
 { number: 38, symbol: "Sr", name: "Strontium" },
 { number: 39, symbol: "Y", name: "Yttrium" },
 { number: 40, symbol: "Zr", name: "Zirconium" },
 { number: 41, symbol: "Nb", name: "Niobium" },
 { number: 42, symbol: "Mo", name: "Molybdenum" },
 { number: 43, symbol: "Tc", name: "Technetium" },
 { number: 44, symbol: "Ru", name: "Ruthenium" },
 { number: 45, symbol: "Rh", name: "Rhodium" },
 { number: 46, symbol: "Pd", name: "Palladium" },
 { number: 47, symbol: "Ag", name: "Silver" },
 { number: 48, symbol: "Cd", name: "Cadmium" },
 { number: 49, symbol: "In", name: "Indium" },
 { number: 50, symbol: "Sn", name: "Tin" },
 { number: 51, symbol: "Sb", name: "Antimony" },
 { number: 52, symbol: "Te", name: "Tellurium" },
 { number: 53, symbol: "I", name: "Iodine" },
 { number: 54, symbol: "Xe", name: "Xenon" },
 { number: 55, symbol: "Cs", name: "Cesium" },
 { number: 56, symbol: "Ba", name: "Barium" },
 { number: 57, symbol: "La", name: "Lanthanum" },
 { number: 58, symbol: "Ce", name: "Cerium" },
 { number: 59, symbol: "Pr", name: "Praseodymium" },
 { number: 60, symbol: "Nd", name: "Neodymium" },
 { number: 61, symbol: "Pm", name: "Promethium" },
 { number: 62, symbol: "Sm", name: "Samarium" },
 { number: 63, symbol: "Eu", name: "Europium" },
 { number: 64, symbol: "Gd", name: "Gadolinium" },
 { number: 65, symbol: "Tb", name: "Terbium" },
 { number: 66, symbol: "Dy", name: "Dysprosium" },
```

```
{ number: 67, symbol: "Ho", name: "Holmium" },
  { number: 68, symbol: "Er", name: "Erbium" },
  { number: 69, symbol: "Tm", name: "Thulium" },
  { number: 70, symbol: "Yb", name: "Ytterbium" },
  { number: 71, symbol: "Lu", name: "Lutetium" },
  { number: 72, symbol: "Hf", name: "Hafnium" },
  { number: 73, symbol: "Ta", name: "Tantalum" },
  { number: 74, symbol: "W", name: "Tungsten" },
  { number: 75, symbol: "Re", name: "Rhenium" },
  { number: 76, symbol: "Os", name: "Osmium" },
  { number: 77, symbol: "Ir", name: "Iridium" },
  { number: 78, symbol: "Pt", name: "Platinum" },
  { number: 79, symbol: "Au", name: "Gold" },
  { number: 80, symbol: "Hg", name: "Mercury" },
  { number: 81, symbol: "Tl", name: "Thallium" },
  { number: 82, symbol: "Pb", name: "Lead" },
  { number: 83, symbol: "Bi", name: "Bismuth" },
  { number: 84, symbol: "Po", name: "Polonium" },
  { number: 85, symbol: "At", name: "Astatine" },
  { number: 86, symbol: "Rn", name: "Radon" },
  { number: 87, symbol: "Fr", name: "Francium" },
  { number: 88, symbol: "Ra", name: "Radium" },
  { number: 89, symbol: "Ac", name: "Actinium" },
  { number: 90, symbol: "Th", name: "Thorium" },
  { number: 91, symbol: "Pa", name: "Protactinium" },
  { number: 92, symbol: "U", name: "Uranium" },
  { number: 93, symbol: "Np", name: "Neptunium" },
  { number: 94, symbol: "Pu", name: "Plutonium" },
  { number: 95, symbol: "Am", name: "Americium" },
  { number: 96, symbol: "Cm", name: "Curium" },
  { number: 97, symbol: "Bk", name: "Berkelium" },
  { number: 98, symbol: "Cf", name: "Californium" },
  { number: 99, symbol: "Es", name: "Einsteinium" },
  { number: 100, symbol: "Fm", name: "Fermium" },
  { number: 101, symbol: "Md", name: "Mendelevium" },
  { number: 102, symbol: "No", name: "Nobelium" },
  { number: 103, symbol: "Lr", name: "Lawrencium" },
  { number: 104, symbol: "Rf", name: "Rutherfordium" },
  { number: 105, symbol: "Db", name: "Dubnium" },
  { number: 106, symbol: "Sg", name: "Seaborgium" },
  { number: 107, symbol: "Bh", name: "Bohrium" },
  { number: 108, symbol: "Hs", name: "Hassium" },
  { number: 109, symbol: "Mt", name: "Meitnerium" },
  { number: 110, symbol: "Ds", name: "Darmstadtium" },
  { number: 111, symbol: "Rg", name: "Roentgenium" },
  { number: 112, symbol: "Cn", name: "Copernicium" },
  { number: 113, symbol: "Nh", name: "Nihonium" },
  { number: 114, symbol: "Fl", name: "Flerovium" },
  { number: 115, symbol: "Mc", name: "Moscovium" },
  { number: 116, symbol: "Lv", name: "Livermorium" },
  { number: 117, symbol: "Ts", name: "Tennessine" },
  { number: 118, symbol: "Og", name: "Oganesson" }
];
```

```
function AtomQuiz() {
  const [question, setQuestion] = useState({});
  const [options, setOptions] = useState([]);
  const [score, setScore] = useState(0);
  const [feedback, setFeedback] = useState('');
  const [questionCount, setQuestionCount] = useState(0);

  useEffect(() => {
    generateQuestion();
  }, []);

  function getRandomElements(count, excludeIndex) {
    const filtered = elements.filter((_, index) => index !== excludeIndex);
    const shuffled = [...filtered].sort(() => 0.5 - Math.random());
    return shuffled.slice(0, count);
  }

  function generateQuestion() {
    const correctIndex = Math.floor(Math.random() * elements.length);
    const correctElement = elements[correctIndex];
    const isSymbolQuestion = Math.random() < 0.5;
    const wrongOptions = getRandomElements(3, correctIndex);
    const answers = [...wrongOptions, correctElement].sort(() => 0.5 - Math.random());

    const questionData = {
      type: isSymbolQuestion ? 'symbol' : 'number',
      question: isSymbolQuestion
        ? `What is the symbol for ${correctElement.name}?`
        : `Which element has atomic number ${correctElement.number}?`,
      correctAnswer: isSymbolQuestion ? correctElement.symbol : correctElement.name,
      answers: answers.map((el) => (isSymbolQuestion ? el.symbol : el.name)),
    };

    setQuestion(questionData);
    setOptions(questionData.answers);
    setFeedback('');
  }
```

```
function handleAnswerClick(answer) {
    if (answer === question.correctAnswer) {
      setScore(score + 1);
      setFeedback('✅ Correct!');
    } else {
      setFeedback(`❌ Wrong! Correct answer: ${question.correctAnswer}`);
    }
    setTimeout(() => {
      setQuestionCount((prev) => prev + 1);
      generateQuestion();
    }, 1200);
  }

  function resetQuiz() {
    setScore(0);
    setQuestionCount(0);
    generateQuestion();
  }

  return (
    <div className="atom-quiz-container">
      <h2>Atom Quiz</h2>
      <p>Score: {score} / {questionCount}</p>
      <div className="question-box">
        <h3>{question.question}</h3>
        <div className="options-grid">
          {options.map((option, index) => (
            <button key={index} className="quiz-option" onClick={() =>
handleAnswerClick(option)}>
              {option}
            </button>
          ))}
        </div>
```

```
<p className="feedback">{feedback}</p>
        <button className="reset-btn" onClick={resetQuiz}>
🔁 Reset Quiz</button>
      </div>
    </div>
  );
}

export default AtomQuiz;
```

```
import React, { useState } from 'react';
import { elements } from '../data/elements';
import '../styles/quiz.css';

function AtomQuiz() {
  const [currentQuestion, setCurrentQuestion] = useState(0);
  const [userAnswer, setUserAnswer] = useState('');
  const [score, setScore] = useState(0);
  const [completed, setCompleted] = useState(false);

  const questions = elements.slice(0, 10).map((el) => ({
    question: `What is the atomic number of ${el.name}?`,
    answer: el.number.toString(),
    options: shuffle([
      el.number.toString(),
      (el.number + 1).toString(),
      (el.number + 2).toString(),
      (el.number - 1).toString(),
    ])
  }));

  function shuffle(array) {
    return array.sort(() => Math.random() - 0.5);
  }

  const handleOptionClick = (option) => {
    if (option === questions[currentQuestion].answer) {
      setScore(score + 1);
    }
```

```
const next = currentQuestion + 1;
    if (next < questions.length) {
      setCurrentQuestion(next);
    } else {
      setCompleted(true);
    }
  };

  return (
    <div className="quiz-container">
      <h2>Element Quiz</h2>
      {completed ? (
        <div>
          <h3>Quiz Complete</h3>
          <p>Your Score: {score} / {questions.length}</p>
        </div>
      ) : (
        <div className="question-block">
          <h4>{questions[currentQuestion].question}</h4>
          <div className="options">
            {questions[currentQuestion].options.map((opt, i) => (
              <button key={i} onClick={() => handleOptionClick(opt)} className="quiz-btn">
                {opt}
              </button>
            ))}
          </div>
        </div>
      )}
    </div>
  );
}

export default AtomQuiz;
```

Quiz.css

```css
.quiz-container {
  padding: 1rem;
  background: #202020;
  color: #fff;
  border-radius: 10px;
}
.question-block {
  margin-top: 1rem;
}
.options {
  display: flex;
  flex-wrap: wrap;
  gap: 10px;
  margin-top: 1rem;
}
.quiz-btn {
  background-color: #3949ab;
  color: white;
  padding: 10px 15px;
  border: none;
  border-radius: 6px;
  cursor: pointer;
}
.quiz-btn:hover {
  background-color: #1e88e5;
}
```

ChemicalReaction.jsx

```
import React, { useState } from 'react';
import { elements } from '../data/elements';
import '../styles/reaction.css';

function ChemicalReaction() {
  const [reactants, setReactants] = useState([]);
  const [result, setResult] = useState('');

  const handleReactantSelect = (el) => {
    if (reactants.length >= 2) return;
    setReactants([...reactants, el]);
  };

  const simulateReaction = () => {
    if (reactants.length < 2) return;
    const names = reactants.map((r) => r.symbol).join(' + ');
    const mockResult = `New Compound:
${reactants[0].symbol}${reactants[1].symbol}`;
    setResult(`${names} → ${mockResult}`);
    setReactants([]);
  };
```

```
return (
  <div className="reaction-container">
    <h2>Chemical Reaction Simulator</h2>
    <div className="reactant-selection">
      {elements.slice(0, 30).map((el) => (
        <button
          key={el.number}
          onClick={() => handleReactantSelect(el)}
          className="react-btn"
        >
          {el.symbol}
        </button>
      ))}
    </div>
    <button className="simulate-btn"
onClick={simulateReaction}>Simulate Reaction</button>
    {result && <div className="reaction-result">{result}</div>}
  </div>
 );
}

export default ChemicalReaction;
```

reaction.css

```css
.reaction-container {
  padding: 1rem;
  background: #141414;
  color: white;
  border-radius: 10px;
}
.reactant-selection {
  display: flex;
  flex-wrap: wrap;
  gap: 10px;
  margin-bottom: 1rem;
}
.react-btn {
  background-color: #8e24aa;
  color: white;
  border: none;
  padding: 10px;
  border-radius: 6px;
  cursor: pointer;
}
```

```
.simulate-btn {
  background-color: #43a047;
  padding: 10px 20px;
  border: none;
  border-radius: 6px;
  cursor: pointer;
}
.reaction-result {
  margin-top: 1rem;
  font-size: 18px;
  font-weight: bold;
  color: #ffeb3b;
}
```

MoleculeMixer.jsx

```
import React, { useState } from 'react';
import { elements } from '../data/elements';
import '../styles/moleculeMixer.css';

function MoleculeMixer() {
  const [selectedElements, setSelectedElements] = useState([]);
  const [moleculeName, setMoleculeName] = useState('');
  const [customName, setCustomName] = useState('');
  const [history, setHistory] = useState([]);

  const handleElementClick = (el) => {
    if (selectedElements.length >= 5 || selectedElements.includes(el)) return;
    setSelectedElements([...selectedElements, el]);
  };

  const handleRemove = (el) => {
    setSelectedElements(selectedElements.filter(e => e !== el));
  };

  const generateFormula = () => {
    const counts = {};
    selectedElements.forEach(el => {
      counts[el.symbol] = (counts[el.symbol] || 0) + 1;
    });
    return Object.entries(counts).map(([sym, count]) => `${sym}${count > 1 ? count : ''}`).join('');
  };

  const handleMix = () => {
    if (selectedElements.length === 0) return;
    const formula = generateFormula();
    const name = customName ? customName : `Mol-${formula}`;
    setMoleculeName(name);
    setHistory([{ name, formula, time: new Date().toLocaleTimeString() }, ...history.slice(0, 4)]);
    setSelectedElements([]);
    setCustomName('');
  };
```

```
const handleReset = () => {
    setSelectedElements([]);
    setMoleculeName('');
    setCustomName('');
  };

  return (
    <div className="mixer-container">
      <h2 className="title">Molecule Mixer</h2>

      <div className="selected-elements">
        {selectedElements.map((e, i) => (
          <span key={i} className="chip" onClick={() => handleRemove(e)}>
            {e.symbol}
          </span>
        ))}
      </div>

      <input
        type="text"
        value={customName}
        onChange={(e) => setCustomName(e.target.value)}
        placeholder="Custom molecule name"
        className="custom-input"
      />

      <div className="actions">
        <button onClick={handleMix} className="btn mix">Mix</button>
        <button onClick={handleReset} className="btn reset">Reset</button>
      </div>

      {moleculeName && (
        <div className="result pulse">
          <h3>Created Molecule:</h3>
          <p>{moleculeName}</p>
          <p>Formula: {generateFormula()}</p>
        </div>
      )}

      <h3>All Elements</h3>
      <div className="elements-grid">
        {elements.map((el) => (
          <button
            key={el.number}
            onClick={() => handleElementClick(el)}
            className={`element-btn ${selectedElements.includes(el) ? 'selected' : ''}`}
          >
            {el.symbol}
          </button>
        ))}
      </div>
```

```
<div className="history">
    <h3>Recent Molecules</h3>
    <ul>
      {history.map((h, i) => (
        <li key={i}>
          {h.name} ({h.formula}) - {h.time}
        </li>
      ))}
    </ul>
  </div>
 </div>
 );
}

export default MoleculeMixer;
```

moleculeMixer.css

```
.mixer-container {
  padding: 2rem;
  background: #1e1e2e;
  color: #ffffff;
  font-family: sans-serif;
}
.title {
  text-align: center;
  font-size: 28px;
}
.elements-grid {
  display: grid;
  grid-template-columns: repeat(auto-fill, minmax(40px, 1fr));
  gap: 8px;
  margin-top: 20px;
}
.element-btn {
  background: #444;
  border: none;
  color: white;
  padding: 8px;
  border-radius: 5px;
  cursor: pointer;
  transition: transform 0.2s;
}
.element-btn:hover {
  transform: scale(1.1);
}
.element-btn.selected {
  background: #6a0dad;
}
.selected-elements {
  display: flex;
  gap: 10px;
  margin-bottom: 10px;
}
```

```css
.chip {
  background: #333;
  padding: 6px 10px;
  border-radius: 20px;
  cursor: pointer;
}
.custom-input {
  width: 100%;
  padding: 10px;
  margin-bottom: 10px;
  background: #292929;
  color: white;
  border: none;
}
.actions {
  display: flex;
  gap: 10px;
  margin-bottom: 20px;
}
.btn {
  padding: 10px 15px;
  cursor: pointer;
  background: #3f51b5;
  color: white;
  border: none;
  border-radius: 5px;
}
.btn.reset {
  background: #d32f2f;
}
.result {
  background: #212121;
  padding: 1rem;
  margin: 1rem 0;
  border-radius: 10px;
}
.pulse {
  animation: pulse 1.5s ease-out;
}
@keyframes pulse {
  0% { transform: scale(0.9); opacity: 0.7; }
  100% { transform: scale(1); opacity: 1; }
}
```

```css
.history {
  margin-top: 30px;
}
.history ul {
  list-style: none;
  padding: 0;
}
.history li {
  padding: 5px 0;
  border-bottom: 1px solid #333;
}
```

ScoreBoard.jsx

```
import React, { useEffect, useState } from 'react';
import axios from 'axios';
import '../styles/scoreboard.css';

function ScoreBoard() {
  const [scores, setScores] = useState([]);
  const [username, setUsername] = useState('');
  const [score, setScore] = useState('');
  const [submitted, setSubmitted] = useState(false);

  useEffect(() => {
    fetchScores();
  }, []);

  const fetchScores = async () => {
    try {
      const response = await axios.get('http://localhost:5000/scores');
      setScores(response.data);
    } catch (error) {
      console.error('Failed to fetch scores');
    }
  };

  const handleSubmit = async (e) => {
    e.preventDefault();
    if (!username || !score) return;
    try {
      await axios.post('http://localhost:5000/scores', { username, score: parseInt(score) });
      setSubmitted(true);
      setUsername('');
      setScore('');
      fetchScores();
    } catch (error) {
      console.error('Failed to submit score');
    }
  };

  return (
    <div className="scoreboard-container">
      <h2>Top Scores</h2>
      <form className="score-form" onSubmit={handleSubmit}>
        <input
          type="text"
          placeholder="Your Name"
          value={username}
          onChange={(e) => setUsername(e.target.value)}
        />
```

```
<input
        type="number"
        placeholder="Your Score"
        value={score}
        onChange={(e) => setScore(e.target.value)}
      />
      <button type="submit">Submit Score</button>
    </form>
    {submitted && <p className="success">Score
submitted successfully</p>}
    <ul className="score-list">
      {scores.map((item, index) => (
        <li key={index}>
          <span className="user">{item.username}</span>
          <span className="score">{item.score}</span>
        </li>
      ))}
    </ul>
  </div>
  );
}

export default ScoreBoard;
```

scoreboard.css

```
.scoreboard-container {
  padding: 2rem;
  background: #121212;
  color: white;
  font-family: 'Segoe UI', sans-serif;
}

.score-form {
  display: flex;
  gap: 10px;
  margin-bottom: 20px;
  flex-wrap: wrap;
}

.score-form input {
  padding: 10px;
  background: #1f1f1f;
  border: 1px solid #444;
  color: white;
  border-radius: 5px;
}

.score-form button {
  padding: 10px 15px;
  background: #009688;
  border: none;
  border-radius: 5px;
  color: white;
  cursor: pointer;
}
```

```css
.success {
  color: #4caf50;
  font-weight: bold;
  margin-bottom: 10px;
}

.score-list {
  list-style: none;
  padding: 0;
  margin: 0;
}

.score-list li {
  display: flex;
  justify-content: space-between;
  background: #1e1e1e;
  padding: 10px;
  margin-bottom: 5px;
  border-radius: 5px;
}

.user {
  font-weight: bold;
  color: #90caf9;
}

.score {
  font-weight: bold;
  color: #ffeb3b;
}
```

main.css

```
import './styles/main.css';
* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
}

html, body {
  font-family: 'Segoe UI', sans-serif;
  background-color: #0f0f0f;
  color: #ffffff;
  height: 100%;
  overflow-x: hidden;
}

#root {
  padding: 1rem;
  min-height: 100vh;
  display: flex;
  flex-direction: column;
}

h1, h2, h3, h4, h5 {
  color: #ffffff;
  margin-bottom: 1rem;
}

a {
  color: #81d4fa;
  text-decoration: none;
}

a:hover {
  color: #29b6f6;
  text-decoration: underline;
}
```

```
button {
  background-color: #00695c;
  color: white;
  border: none;
  padding: 10px 20px;
  font-size: 16px;
  border-radius: 8px;
  cursor: pointer;
  transition: all 0.2s ease-in-out;
}

button:hover {
  background-color: #004d40;
  transform: scale(1.03);
}

input, select, textarea {
  background-color: #1c1c1c;
  color: white;
  border: 1px solid #555;
  padding: 10px;
  border-radius: 6px;
  font-size: 15px;
  width: 100%;
}

.container {
  max-width: 1200px;
  margin: 0 auto;
  padding: 1rem;
}

.flex {
  display: flex;
  gap: 1rem;
}
```

```css
.flex-center {
  display: flex;
  justify-content: center;
  align-items: center;
}

.grid {
  display: grid;
  gap: 1rem;
}

.grid-cols-2 {
  grid-template-columns: repeat(2, 1fr);
}

.grid-cols-3 {
  grid-template-columns: repeat(3, 1fr);
}

.grid-cols-4 {
  grid-template-columns: repeat(4, 1fr);
}

.card {
  background-color: #1e1e1e;
  padding: 1.5rem;
  border-radius: 10px;
  box-shadow: 0 4px 8px rgba(0, 255, 255, 0.1);
  transition: transform 0.2s ease;
}
```

```css
.card:hover {
  transform: scale(1.02);
}

.text-center {
  text-align: center;
}

.text-right {
  text-align: right;
}

.text-left {
  text-align: left;
}

.rounded {
  border-radius: 10px;
}

.mt-1 { margin-top: 0.25rem; }
.mt-2 { margin-top: 0.5rem; }
.mt-3 { margin-top: 1rem; }

.mb-1 { margin-bottom: 0.25rem; }
.mb-2 { margin-bottom: 0.5rem; }
.mb-3 { margin-bottom: 1rem; }

.p-1 { padding: 0.25rem; }
.p-2 { padding: 0.5rem; }
.p-3 { padding: 1rem; }

.fade-in {
  animation: fadein 0.8s ease-out forwards;
}
```

```css
@keyframes fadein {
  0% { opacity: 0; transform: translateY(20px); }
  100% { opacity: 1; transform: translateY(0); }
}

.shadow {
  box-shadow: 0px 2px 5px rgba(255, 255, 255, 0.05);
}

.bg-dark {
  background-color: #121212;
}

.bg-light {
  background-color: #f4f4f4;
  color: #222;
}

.text-bangla {
  font-family: 'Noto Sans Bengali', sans-serif;
  font-size: 16px;
}

.lang-toggle {
  position: fixed;
  top: 10px;
  right: 10px;
  background-color: #37474f;
  padding: 6px 12px;
  color: white;
  border-radius: 4px;
  cursor: pointer;
}
```

server.js

```
const express = require('express');
const cors = require('cors');
const mongoose = require('mongoose');

const app = express();
const PORT = process.env.PORT || 5000;

app.use(cors());
app.use(express.json());

mongoose.connect('YOUR_MONGO_ATLAS_URI', {
  useNewUrlParser: true,
  useUnifiedTopology: true
});

const db = mongoose.connection;
db.on('error', console.error.bind(console, 'MongoDB error:'));
db.once('open', () => console.log('Connected to MongoDB Atlas'));

const scoreSchema = new mongoose.Schema({
  username: { type: String, required: true },
  score: { type: Number, required: true },
  date: { type: Date, default: Date.now }
});
```

```
const Score = mongoose.model('Score', scoreSchema);

app.get('/scores', async (req, res) => {
  try {
    const scores = await Score.find().sort({ score: -1 }).limit(10);
    res.json(scores);
  } catch (error) {
    res.status(500).json({ error: 'Failed to fetch scores' });
  }
});

app.post('/scores', async (req, res) => {
  const { username, score } = req.body;
  if (!username || typeof score !== 'number') {
    return res.status(400).json({ error: 'Invalid input' });
  }

  try {
    const newScore = new Score({ username, score });
    await newScore.save();
    res.status(201).json({ message: 'Score saved' });
  } catch (error) {
    res.status(500).json({ error: 'Failed to save score' });
  }
});

app.listen(PORT, () => {
  console.log(`Server running on port ${PORT}`);
});
```

Thank You